

DW_norm_rnd

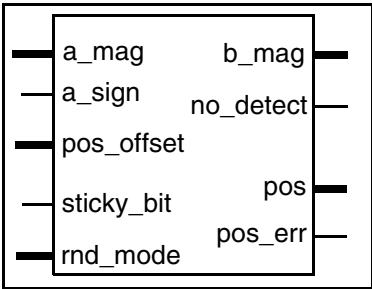
Normalization and rounding

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Features and Benefits

- Parameterized word lengths
- Parameterized search window
- Exponent calculation useful for floating-point numbers
- Implements the most used rounding modes
- Provides status indications for exceptional cases
- DesignWare datapath generator is employed for better timing and area

Revision History



Description

DW_norm_rnd is a general-purpose normalization and rounding module for a value represented in the sign-and-magnitude number system (a_sign, a_mag). The value represented in this system corresponds to $(-1)^{a_sign} a_mag$. The magnitude of the number is a positive fixed-point value in the format $a = (a_0.a_1 a_2 a_3 a_4 a_5 \dots a_{a_width-1})$, where a_i represents a bit. This means that a_mag has 1 integer bit and $(a_width - 1)$ fractional bits.

The normalization process consists in generating an output in the range $1 \leq b_mag < 2$ (the output bit-vector has a 1 in the MS bit position) when there is a 1 bit in the search window provided as a parameter.

The rounding is done using a method controlled by one of the component inputs. Output `pos` carries information about the position of binary point (exponent), and it is affected by the normalization shifts performed on the main input (a_mag). A list of pins for this component is provided in [Table 1-1](#).

The number of bit positions in a_mag shifted during initial normalization and post-rounding normalization (n bit positions) is reflected in the value of `pos`. This output corresponds to $(pos_offset + n)$ when parameter $exp_ctr = 0$, or to $(pos_offset - n)$ when $exp_ctr = 1$. When the result of this operation does not fit in the bit-vector used for `pos` or it is negative, the output `pos_err` is set to 1.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
a_mag	a_width bits	Input	Input data
a_sign	1 bit	Input	■ 0: Positive ■ 1: Negative
pos_offset	exp_width bits	Input	Offset value for the position of the binary point

Table 1-1 Pin Description (Continued)

Pin Name	Width	Direction	Function
sticky_bit	1 bit	Input	Indicates the presence of non-zero bits in fractional bit positions beyond bit $a_mag_{a_width-1}$
rnd_mode	3 bits	Input	Rounding mode 000: Round to nearest even 001: Round towards zero 010: Round to plus infinity 011: Round to minus infinity 100: Round up 101: Round away from zero
no_detect	1 bit	Output	Result of MS 1 bit search in the search window. 0: Bit found 1: Bit not found
pos_err	1 bit	Output	Value provided at output <code>pos</code> cannot fit in an exp_width - bit vector or <code>pos</code> is negative
b_mag	b_width bits	Output	Normalized and rounded output data
pos	exp_width bits	Output	<code>pos_offset</code> combined with the number of bit positions that input <code>a_mag</code> was shifted (n): $exp_ctr = 0 \rightarrow pos_offset + n$ $exp_ctr = 1 \rightarrow pos_offset - n$

Table 1-2 Parameter Description

Parameter	Values	Description
a_width	≥ 2 Default: 16	Word length of <code>a_mag</code>
$srch_wind$	2 to a_width Default: 4	Search window for the MS 1 bit (from left to right, or from bit 0 to $a_width - 1$)
exp_width	≥ 1 Default: 4	Word length of <code>pos_offset</code> and <code>pos</code>
b_width	2 to a_width Default: 10	Word length of <code>b_mag</code>
exp_ctr	0 or 1 Default: 0	Controls computation of the binary point position (<code>pos</code> output)

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW01.DW_NORM_RND_CFG_SIM	Design unit name for VHDL simulation
dw/dw01/src/DW_norm__rnd_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_norm_rnd.v	Verilog simulation model source code

The input `pos_offset` is provided by the user to indicate that pre-shifting of the input operand was already done to adjust it to the required input format. For example, if the actual fixed-point value is (101.0011), the input for the DW_norm_rnd component could be `a_mag` = (0.01010011) and `pos_offset` = (0100) for the parameters `a_width` = 9 and `exp_width` = 4. Observe that the designer can use other input combinations as needed.

The `srch_wind` parameter reduces the complexity of this component when it is known that the MS 1 bit resides in a limited group of MS bit positions. When there is no 1 bit in the search window, the input `a_mag` is shifted to the left by (`srch_wind` - 1) bit positions and `no_detect` is set to 1.

Once the input is normalized, a rounding method is applied to it according to the `rnd_mode` input. Rounding consists in truncating the normalized input vector (`a_norm`) to `b_width` bits and adding 1 or 0 (value called `rnd` in the following equations) to the least-significant bit position of this truncated normal vector (fractional bit $L = a_norm_{b_width - 1}$). In order to make a decision about the value of `rnd`, the rounding procedure makes use of two extra bits called round bit (`R`) and sticky bit (`T`). These bits are also obtained from the normalized vector `a_norm`, and for this reason the user must make sure that enough significant bits are provided to the normalization and rounding unit to allow proper rounding. The round bit corresponds to `a_norm_{b_width}` when `b_width` < `a_width`; otherwise it has value 0. The sticky bit is a combination of input `sticky_bit` and any other bits beyond `a_norm_{b_width}` (value after normalization). The `sticky_bit` input indicates that there are other non-zero bits beyond the fractional bit position `a_width` - 1 in the main input `a_mag`.

Once the bits `R` and `T` are determined, the value of `rnd` used for rounding is generated as:

- RNE: $rnd = R$ and (L or T)
- Rzero: $rnd = 0$ (only truncated value)
- Rpos: $rnd = \text{not } a_sign$ and (R or T)
- Rneg: $rnd = a_sign$ and (R or T)
- Rup: $rnd = R$
- Raway: $rnd = R + T$

The component detects the cases that require post-rounding normalization and adjusts the `pos` output accordingly. For these cases, when `exp_ctr = 0`, the exponent value at output `pos` is decremented by 1, and when `exp_ctr = 1`, the exponent value is incremented by 1.

Related Topics

- [Datapath– Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```

library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW_norm_rnd_inst is
    generic (
        inst_a_width : POSITIVE := 16;
        inst_srch_wind : POSITIVE := 4;
        inst_exp_width : POSITIVE := 4;
        inst_b_width : POSITIVE := 10;
        inst_exp_ctr : INTEGER := 0
    );
    port (
        inst_a_mag : in std_logic_vector(inst_a_width-1 downto 0);
        inst_pos_offset : in std_logic_vector(inst_exp_width-1 downto 0);
        inst_sticky_bit : in std_logic;
        inst_a_sign : in std_logic;
        inst_rnd_mode : in std_logic_vector(2 downto 0);
        pos_err_inst : out std_logic;
        no_detect_inst : out std_logic;
        b_inst : out std_logic_vector(inst_b_width-1 downto 0);
        pos_inst : out std_logic_vector(inst_exp_width-1 downto 0)
    );
end DW_norm_rnd_inst;

architecture inst of DW_norm_rnd_inst is

begin

    -- Instance of DW_norm_rnd
    U1 : DW_norm_rnd
        generic map ( a_width => inst_a_width, srch_wind => inst_srch_wind, exp_width =>
inst_exp_width, b_width => inst_b_width, exp_ctr => inst_exp_ctr )
        port map ( a_mag => inst_a_mag, pos_offset => inst_pos_offset, sticky_bit =>
inst_sticky_bit, a_sign => inst_a_sign, rnd_mode => inst_rnd_mode, pos_err =>
pos_err_inst, no_detect => no_detect_inst, b => b_inst, pos => pos_inst );

end inst;

```

HDL Usage Through Component Instantiation - Verilog

```
module DW_norm_rnd_inst( inst_a_mag, inst_pos_offset, inst_sticky_bit, inst_a_sign,
    inst_rnd_mode,
        pos_err_inst, no_detect_inst, b_inst, pos_inst );

parameter a_width = 16;
parameter srch_wind = 4;
parameter exp_width = 4;
parameter b_width = 10;
parameter exp_ctr = 0;

input [a_width-1 : 0] inst_a_mag;
input [exp_width-1 : 0] inst_pos_offset;
input inst_sticky_bit;
input inst_a_sign;
input [2 : 0] inst_rnd_mode;
output pos_err_inst;
output no_detect_inst;
output [b_width-1 : 0] b_inst;
output [exp_width-1 : 0] pos_inst;

    // Instance of DW_norm_rnd
    DW_norm_rnd #(a_width, srch_wind, exp_width, b_width, exp_ctr)
        U1 ( .a_mag(inst_a_mag), .pos_offset(inst_pos_offset),
            .sticky_bit(inst_sticky_bit), .a_sign(inst_a_sign), .rnd_mode(inst_rnd_mode),
            .pos_err(pos_err_inst), .no_detect(no_detect_inst), .b(b_inst), .pos(pos_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
December 2020	DWBB_202009.2	■ For STAR 3408988, corrected type for parameter exp_ctr in the example in “HDL Usage Through Component Instantiation - VHDL” on page 5
July 2020	DWBB_201912.5	■ Added the exp_ctr parameter to the example “HDL Usage Through Component Instantiation - VHDL” on page 5
January 2019	DWBB_201806.5	■ Updated example in “HDL Usage Through Component Instantiation - VHDL” on page 5 ■ Added this Revision History table and the document links on this page

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